

Skills Summary

Building Bigger Shapes from Smaller Ones

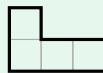
Use this reference while completing your worksheet or when studying.

Skill 1 — Counting the small squares in a built shape

Rule: Small shapes build a bigger one when they leave **no gaps** and **no overlaps**.

To find how much a built shape covers, count its small squares — touch each one once, and count a whole row at a time when the rows are straight.

Example: How many small squares cover this shape?



Step 1. The shape is not a plain rectangle, so I cannot count one row and double it; instead I break it into straight rows and count each row on its own.

Step 2. The bottom row holds 3 small squares and there is 1 more sitting above the left-hand end, and nothing is counted twice.

4 small squares

Try it: How many small squares cover a rectangle with 3 in each row and 2 rows?

check: 6 small squares

Skill 2 — Naming the shape the pieces make

Rule: The built shape gets its own name, and it is usually not the name of the pieces.

Look at the *outside* edge only: four straight sides with square corners is a rectangle, and a rectangle with all four sides the same length is a square.

Example: Two matching triangles are joined along their long edges. What shape do they make, and how much does it cover?



Step 1. To name a built shape I have to ignore the join line and trace only the outside edge, because the pieces inside are not what is being named.

Step 2. The outside edge has four straight sides, square corners, and all four sides the same length, so it is a square; under it lie 2 small squares in each of 2 rows.

a square, covering 4 small squares

Try it: Two small squares are placed side by side. Name the new shape, and say how many small squares it covers.

check: a rectangle, covering 2 small squares

Skill 3 — Comparing two built shapes

Rule: A longer outline does not mean more squares. Count both shapes, then compare the two counts with $<$, $>$, or $=$.

Two shapes that look nothing alike can cover exactly the same amount, because moving squares around adds none and takes none away.

Example: Shape A covers 6 small squares and shape B covers 4. Which is greater?

Step 1. Comparing pictures by eye is what misleads here, so I set the two counts side by side and let the numbers decide.

Step 2. Counting up, 4 is said before 6, so shape B covers less and shape A covers more; the wide end of the symbol faces the bigger count.

$$6 > 4$$

Try it: Shape A covers 5 small squares and so does shape B, although they look different. Which symbol goes between them?

$$5 = 5$$

Skill 4 — Finishing a shape that is not done yet

Rule: To find how many more pieces a build still needs, do not add them one at a time.

Count what the finished outline would hold, count what is already there, and take one from the other: still needed = whole – already placed.

Example: A rectangle will hold 6 small squares. So far 2 are in place. How many more are needed?

Step 1. Placing squares one by one is slow and easy to miscount, so I work with the two totals instead and take the smaller from the bigger.

Step 2. The finished rectangle holds 6 and 2 are already there, so what is missing is $6 - 2$.

4 more small squares

Try it: A square will hold 9 small squares and 6 are in place. How many more are needed?

$$9 - 6 = 3$$